

IAM Quotas in GCP

IAM Quotas in Google Cloud Platform (GCP): Explanation, Use Cases, and Benefits

Definition:

IAM (Identity and Access Management) quotas in Google Cloud Platform refer to the limits imposed by Google Cloud on the number of IAM resources and API operations that can be used within a given project, organization, or across the platform. These quotas are part of GCP's quota management system, which exists to protect users from accidental overuse and to ensure fair usage across customers.

IAM quotas specifically govern:

- The number of service accounts per project
- The number of roles per project or organization
- The number of IAM policy bindings
- The number of permission checks per second
- The rate of IAM API calls (e.g., `setIamPolicy`, `getIamPolicy`, `testIamPermissions`)

Use Cases:

1. **Managing Access at Scale:**

Large enterprises often manage thousands of users, service accounts, and roles. IAM quotas help ensure that their access control infrastructure doesn't overwhelm GCP resources.

2. **Automation and DevOps:**

Automated pipelines and Infrastructure-as-Code tools (e.g., Terraform, Deployment Manager) interact heavily with IAM policies. Quotas prevent runaway scripts from unintentionally flooding the IAM APIs, which could degrade performance or cause outages.

3. **Custom Role Creation:**

Teams may need to define custom IAM roles for fine-grained access control. Quotas on custom roles ensure that role definitions remain manageable and do not overload the IAM system.

4. **High-Frequency Permissions Checks:**

Applications with dynamic access checks (e.g., multi-tenant SaaS platforms) may hit permission-checking limits. Monitoring quotas helps architects optimize IAM usage and caching strategies.

5. **Security Auditing and Monitoring:**

Organizations performing frequent security reviews and audits might generate a high volume of IAM-related API calls. Quotas help detect anomalies or abuse by setting usage ceilings.

Benefits:

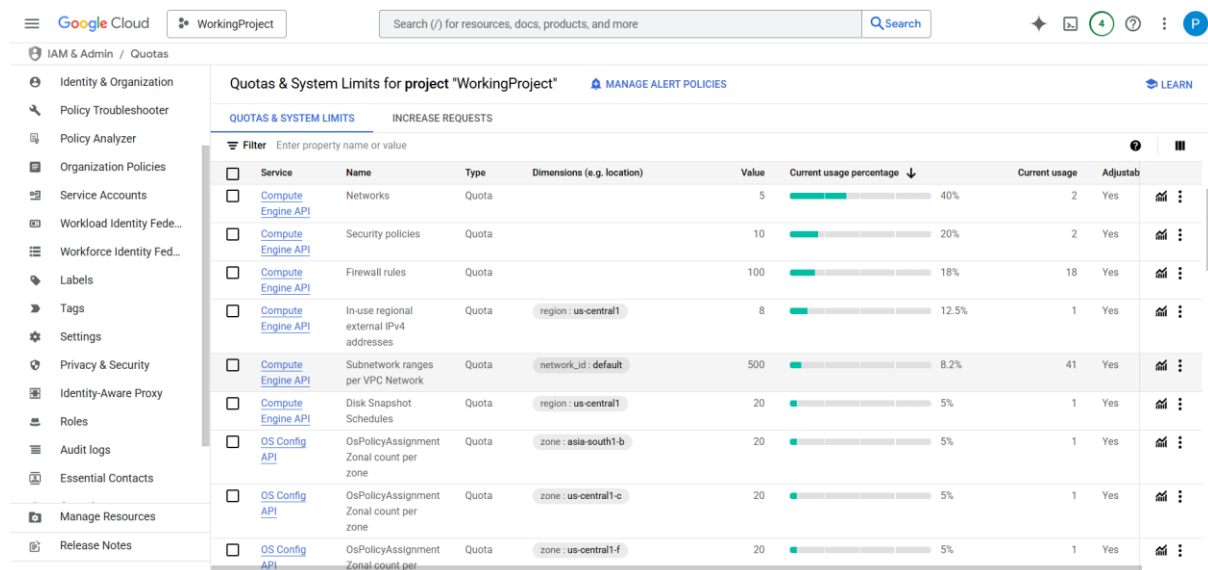
- **System Reliability:** Quotas protect the underlying IAM infrastructure from excessive load, ensuring consistent performance across GCP for all users.
- **Fair Resource Allocation:** They help prevent any one user or project from monopolizing shared resources, which is especially important in multi-tenant cloud environments.
- **Cost Management and Abuse Prevention:** By capping usage, quotas can help detect and prevent unintentional or malicious actions that could incur high costs or cause configuration errors.
- **Scalability Planning:** Quotas highlight system limitations, allowing organizations to plan growth, request quota increases proactively, and align architectural decisions with platform constraints.
- **Security Posture Management:** Quotas on the number of policy bindings or service accounts enforce disciplined and secure access control models, reducing the surface area for misconfigurations.

Conclusion:

IAM quotas in GCP are critical for maintaining security, performance, and governance in cloud environments. They support a scalable and manageable approach to identity and access control by enforcing limits on key IAM operations and entities. Understanding and monitoring these quotas is essential for secure and efficient cloud operations.

To begin with the Lab

1. From the Google Cloud Console, search for IAM and from the left pane navigate to Quotas and System Limits.
2. Here you can see that we have the services and the values or limits for them. We can also see the Current usage and position.

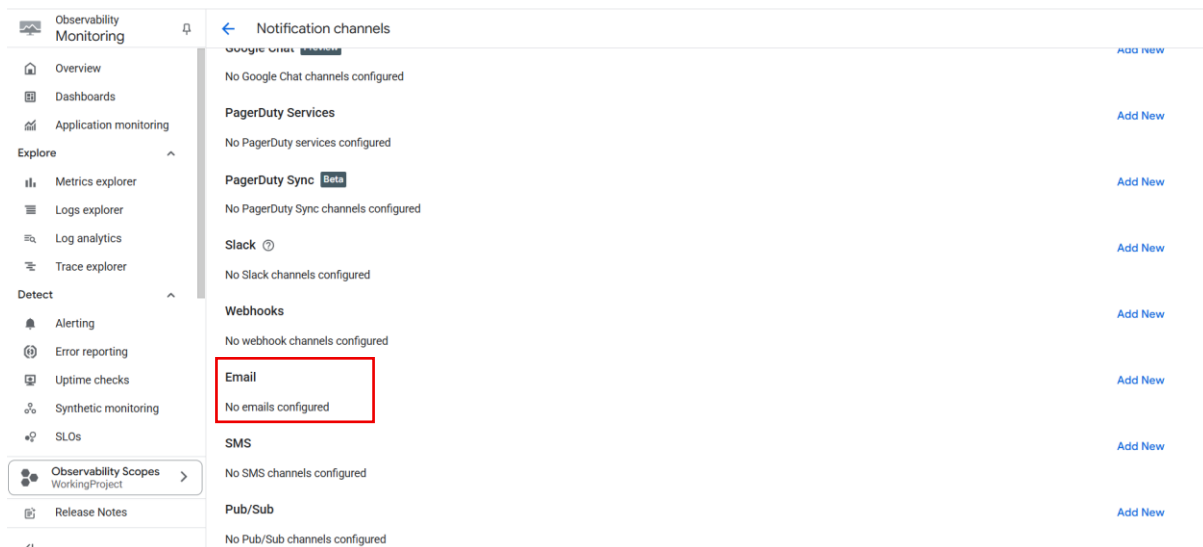


The screenshot shows the Google Cloud IAM & Admin console. The left sidebar contains navigation links for Identity & Organization, Policy Troubleshooter, Policy Analyzer, Organization Policies, Service Accounts, Workload Identity Federation, Workforce Identity Federation, Labels, Tags, Settings, Privacy & Security, Identity-Aware Proxy, Roles, Audit logs, Essential Contacts, Manage Resources, and Release Notes. The main content area is titled 'Quotas & System Limits for project "WorkingProject"'. It includes a search bar and a 'MANAGE ALERT POLICIES' link. Below the title, there are tabs for 'QUOTAS & SYSTEM LIMITS' and 'INCREASE REQUESTS'. A filter bar is present above a table of quotas. The table has columns for Service, Name, Type, Dimensions (e.g. location), Value, Current usage percentage, Current usage, and Adjustab. The table lists various quotas for services like Compute Engine API, OS Config API, and Network API.

Service	Name	Type	Dimensions (e.g. location)	Value	Current usage percentage	Current usage	Adjustab
Compute Engine API	Networks	Quota		5	40%	2	Yes
Compute Engine API	Security policies	Quota		10	20%	2	Yes
Compute Engine API	Firewall rules	Quota		100	18%	18	Yes
Compute Engine API	In-use regional external IPv4 addresses	Quota	region: us-central1	8	12.5%	1	Yes
Compute Engine API	Subnetwork ranges per VPC Network	Quota	network_id: default	500	8.2%	41	Yes
Compute Engine API	Disk Snapshot Schedules	Quota	region: us-central1	20	5%	1	Yes
OS Config API	OsPolicyAssignment Zonal count per zone	Quota	zone: asia-south1-b	20	5%	1	Yes
OS Config API	OsPolicyAssignment Zonal count per zone	Quota	zone: us-central1-c	20	5%	1	Yes
OS Config API	OsPolicyAssignment Zonal count per zone	Quota	zone: us-central1-f	20	5%	1	Yes

3. Now we can create an alert for the quota, but before that, we need to create a notification channel.

- Search and navigate to the notification channel, then choose to create a channel using Email.



- Give an email address, then a display name. Click on Save.

Create Email Channel

Email addresses can be set to receive notifications from your alerting when a new incident is created.

Email Address *
pulkit.k2711@gmail.com

Display Name *
Quotas Alert

Cancel Save

- Now, come back to quotas choose any service you want to create a usage alert for. Then just click on the three dots and choose Create option.

Filter Enter property name or value									
☐	Service	Name	Type	Dimensions (e.g. location)	Value	Current usage percentage ↓	Current usage	Adjustab	
☐	Compute Engine API	Networks	Quota		5	40%	2	Yes	
☐	Compute Engine API	Security policies	Quota		10	20%			
☐	Compute Engine API	Firewall rules	Quota		100	18%	18	Yes	

Create usage alert

Edit quota

- Then you just need to choose your notification channel and click on create.

Configure notifications Recommended

Select where to receive notifications when incidents are triggered. Multiple channels are recommended for redundancy.

☒ Use notification channel

Notification Channels

Quotas Alert

CREATE

CANCEL

8. On the top of the page, you will manage alert policies option. Open it.

Quotas & System Limits for project "WorkingProject"

[MANAGE ALERT POLICIES](#)

QUOTAS & SYSTEM LIMITS

INCREASE REQUESTS

9. You will see a policy here and inside it you can view the other options

[Policies](#) [+ Create policy](#)

[Filter](#) Filter policies

[Filter](#)

Display name	Type ?	Last modified by	Last modified on ↓	Labels	Enabled	
Quota usage - Compute Engine API - Networks - compute.googleapis.com/networks	PromQL	pulkit.k2711@gmail.com	May 28, 2025		☒ on	⋮